E-Commerce Sales Data Analysis and Dashboard Development

Transforming Raw Transactions into Strategic Business Intelligence

Project Details

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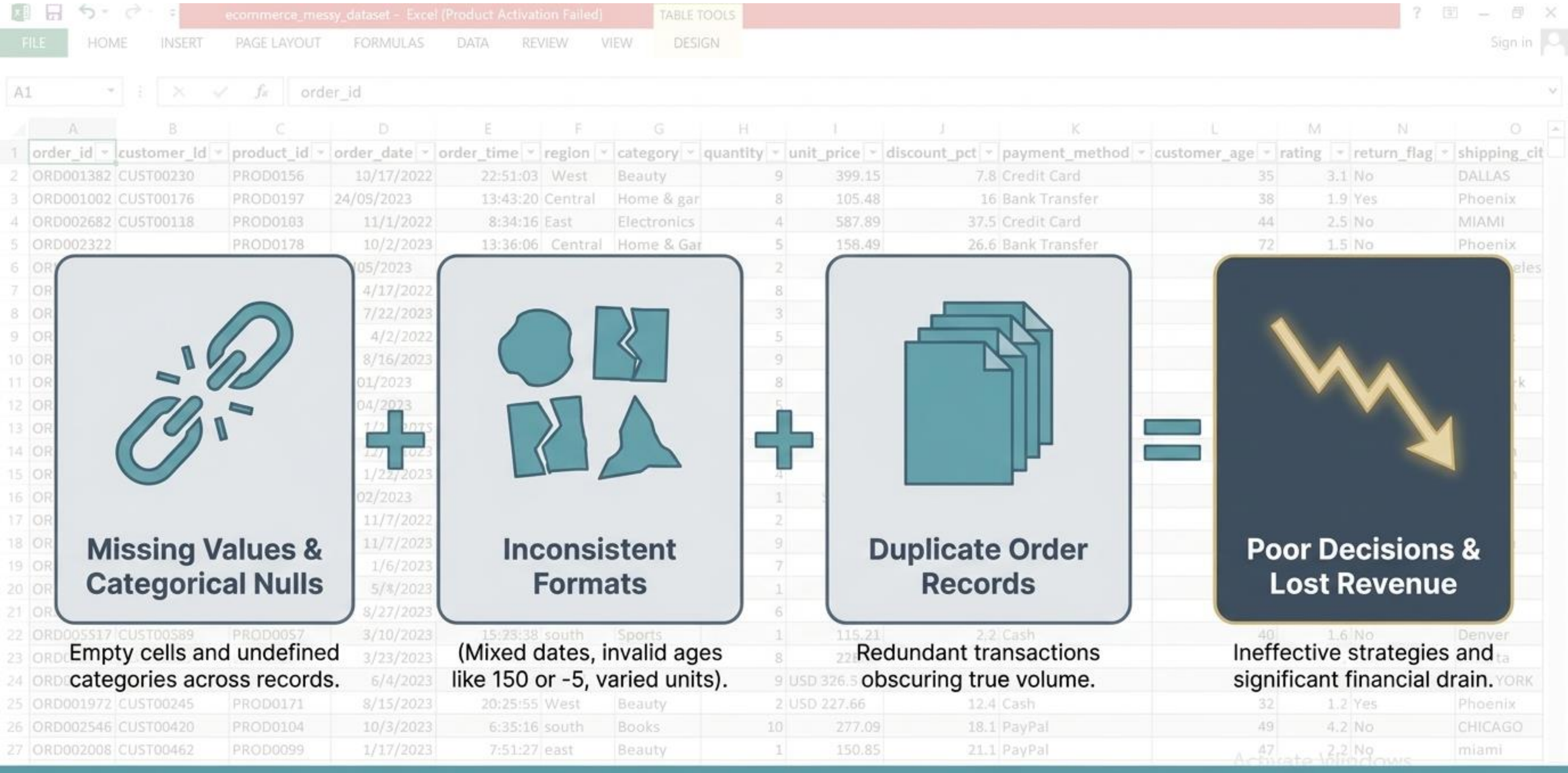
Cohort: 3MTT C2 Advanced Data Analysis 102



An E-commerce business collects vast amounts of data, but without structured transformation, raw transactions obscure critical insights and mask operational inefficiencies.

Unified Business Intelligence

1. Cleanse & Structure

Transform the messy e-commerce dataset into a validated, structured format, correcting anomalies and ensuring data integrity.

2. Extract & Analyze

Identify hidden patterns across dimensions (City, Age, Time) to understand what drives revenue and customer retention.

3. Build & Empower

Develop an interactive, self-updating dashboard to answer critical business questions and guide executive decision-making.

Key Business Questions Addressed

- Which regions generate the most revenue?
- What are the peak sales periods?
- How do discounts affect overall revenue?
- What customer segments drive the most sales?

Raw Data → Business Intelligence

Phase 1: Exploration & Triage

Microsoft Excel

Role: Initial data exploration, structural assessment, and identifying duplicate patterns.

Phase 2: Pre-Processing & Scripting

Python (Pandas)

Role: Programmatic data cleaning, handling nulls, and resolving complex inconsistencies.

Phase 3: Transformation (ETL)

Power Query (M Code)

Role: Building scalable data transformation steps and custom conditional logic.

Phase 4: Modeling & Visualization

DAX & Power BI

Role: Creating dynamic calculated measures (Total Revenue, Order Quantities) and designing the interactive executive dashboard.

From Chaos to Clarity

1. Cleanse

- Imputed missing numerical values via mean/median.
- Standardized categorical text (e.g., shipping cities).
- Corrected invalid age values (e.g., ages 150 → 15, -5 → 50).

2. Transform

- Combined order_date and order_time into unified datetime logic.
- Segmented customers into distinct age groups (Teen, Young Adult, Adult, Senior).
- Engineered financial columns: Revenue = Quantity × Unit Price × (1 - Discount%).

3. Model

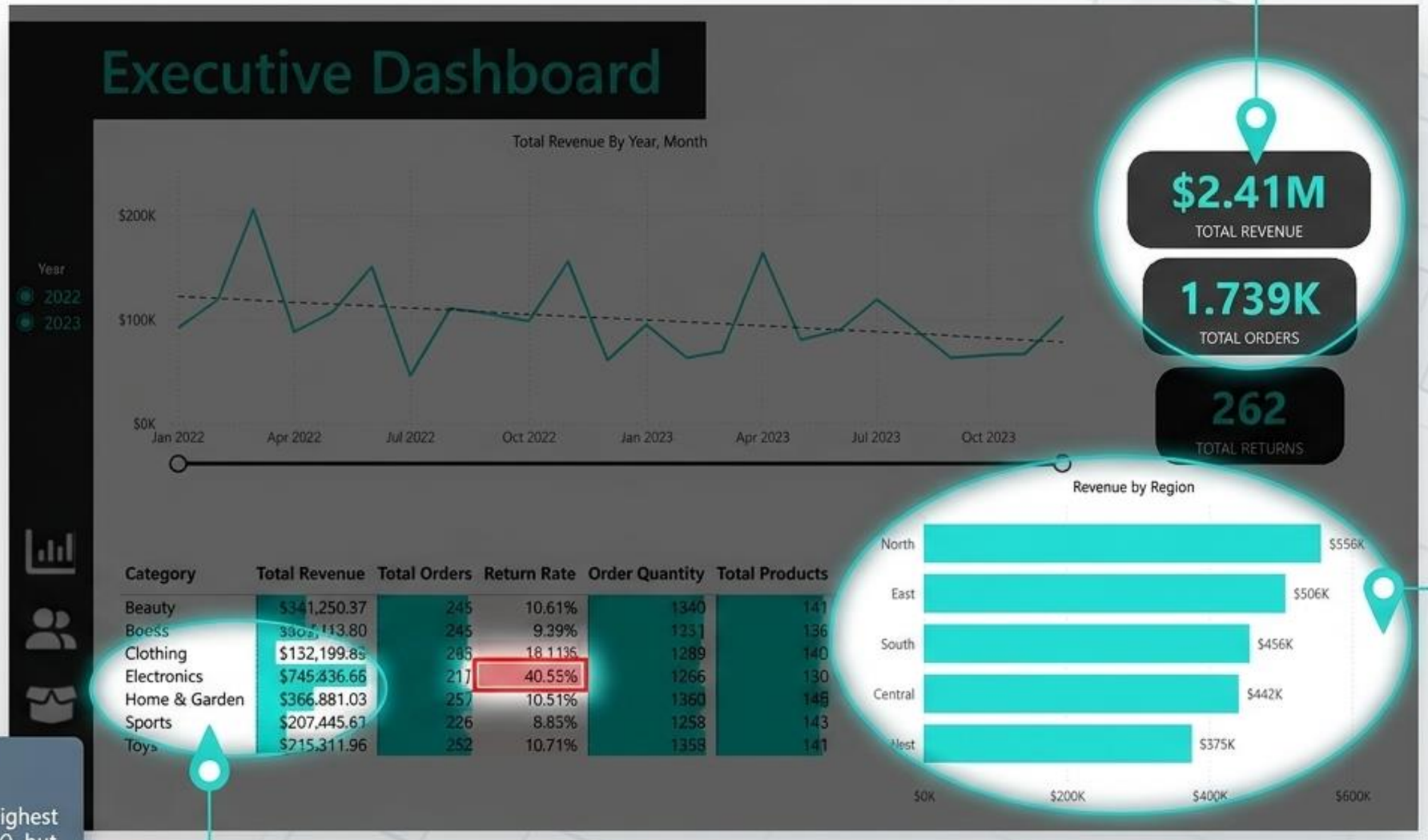
- Built relational models and advanced DAX measures.
- Created dynamic tracking for Return Rate = DIVIDE(SUM(Normalized_Return_Flag), COUNT(Normalized_Return_Flag)).

L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
customer_age	rating	return_flag	shipping_city	order_date_clean	order_datetime	year	month	day	hour	normalized_return_flag	revenue	age_group	
35	3.9	No	Dallas	10/17/2077	10/17/2023 22:51	2022	10	17	27	0	3312.1467	Young Adult	
35	1.8	Yes	Phoenix	9/24/2023	9/24/2023 13:43	2023	9	24	13	1	708.6256	Adult	
44	2.5	No	Miami	10/1/2023	11//2023 13:34	2023	11	1	8	0	1469.725	Adult	
72	1.2	No	Phoenix	10/2/2023	10/2/2023 13:36	2023	10	2	13	0	708.0893	Senior	
52	3.9	No	Los Angeles	5/14/2023	5/14/2023 19:32	2023	5	14	19	0	68.8	Middle Age	
37	3.5	No	Miami	4/12/2022	4/12/2023 19:32	2023	4	12	19	0	1547.7836	Adult	
47	2.7	No	Denver	7/22/2023	7/22/2023 19:32	2023	7	22	18	0	970.15464	Adult	
55	2.9	No	Atlantax	8/6/2022	8/5/2022 16:51	2023	4	2	16	0	189.74991	Middle Age	
24	1.1	No	Miami	7/12/2023	7/12/2023 13:10	2023	7	12	12	0	276.0121	Teen	
70	3.5	No	Houston	1/12/2023	12/2/2023 12:22	2023	12	22	13	0	1548.60852	Young Adult	
76	3.6	No	Seattle	1/17/2022	11/7/2022 16:19	2023	1	7	18	0	413.19398	Young Adult	
32	3.2	No	Houston	1/6/2023	1/5/2023 12:02	2023	11	11	16	0	1304.04271	Senior	
50	4.0	No	Denver	9/37/2023	3/23/2023 17:58	2023	8	27	15	0	1535.04582	Young Adult	
49	4.9	No	Denver	3/18/2023	3/6/2022 22:39	2023	3	10	6	0	1985.2671	Adult	
47	4.7	No	Miami	1/17/2022	1/1/2023 75:51	2023	1	17	17	0	119.00265	Adult	
59	1.9	No	Phoenix	3/16/2022	3/16/2023 75:12	2023	3	16	15	0	1005.66	Middle Age	

Final Transformed Dataset Ready for BI Integration

The Bottom Line

Robust overall sales volume across the 2022-2023 period.



Category	Total Revenue	Total Orders	Return Rate	Order Quantity	Total Products
Beauty	$341,250.37	245	10.61%	1340	141
Boess	[illegible]	245	9.39%	1231	136
Clothing	$132,199.85	265	18.113%	1289	140
Electronics	$745,436.66	217	40.55%	1266	130
Home & Garden	$366,881.03	257	10.51%	1360	145
Sports	$207,445.61	226	8.85%	1258	143
Toys	$215,311.96	252	10.71%	1358	141

Category Risk

Electronics drive the highest overall revenue ($749K), but suffer from a severely elevated return rate of 40.55%.

Regional Parity

Highly balanced regional performance. North leads slightly at $556K, but all regions contribute evenly down to the West at $375K.

Total Base: 563 Unique Customers

Total Customers by City & Age Group

Shipping_City	Adult	Middle Age	Senior	Teen	Young Adult	Total
Atlanta	47	57	11	3	40	143
Chicago	51	57	24	5	38	133
Dallas	39	50	15	2	49	142
Denver	50	53	14	6	35	145
Houston	55	53	9	5	41	150
Los Angeles	41	51	15	6	47	179
Miami	55	43	15	7	45	146
New York	41	63	13	8	44	167
Phoenix	52	59	22	3	56	149
Seattle	52	41	11	4	51	143
Total	335	351	136	45	309	563

Order Quantity by City & Age Group

Shipping_City	Adult	Middle Age	Senior	Teen	Young Adult	Total
Atlanta	297	315	61	12	229	914
Chicago	261	323	142	37	187	930
Dallas	228	267	90	18	251	854
Denver	329	277	78	42	193	919
Houston	328	270	52	35	235	915
Los Angeles	237	313	66	35	276	976
Miami	230	360	65	49	251	988
New York	230	360	86	51	381	968
Phoenix	320	367	135	28	334	1184
Seattle	291	232	63	16	299	901
Total	2851	2983	869	323	2516	9516

The Core Persona

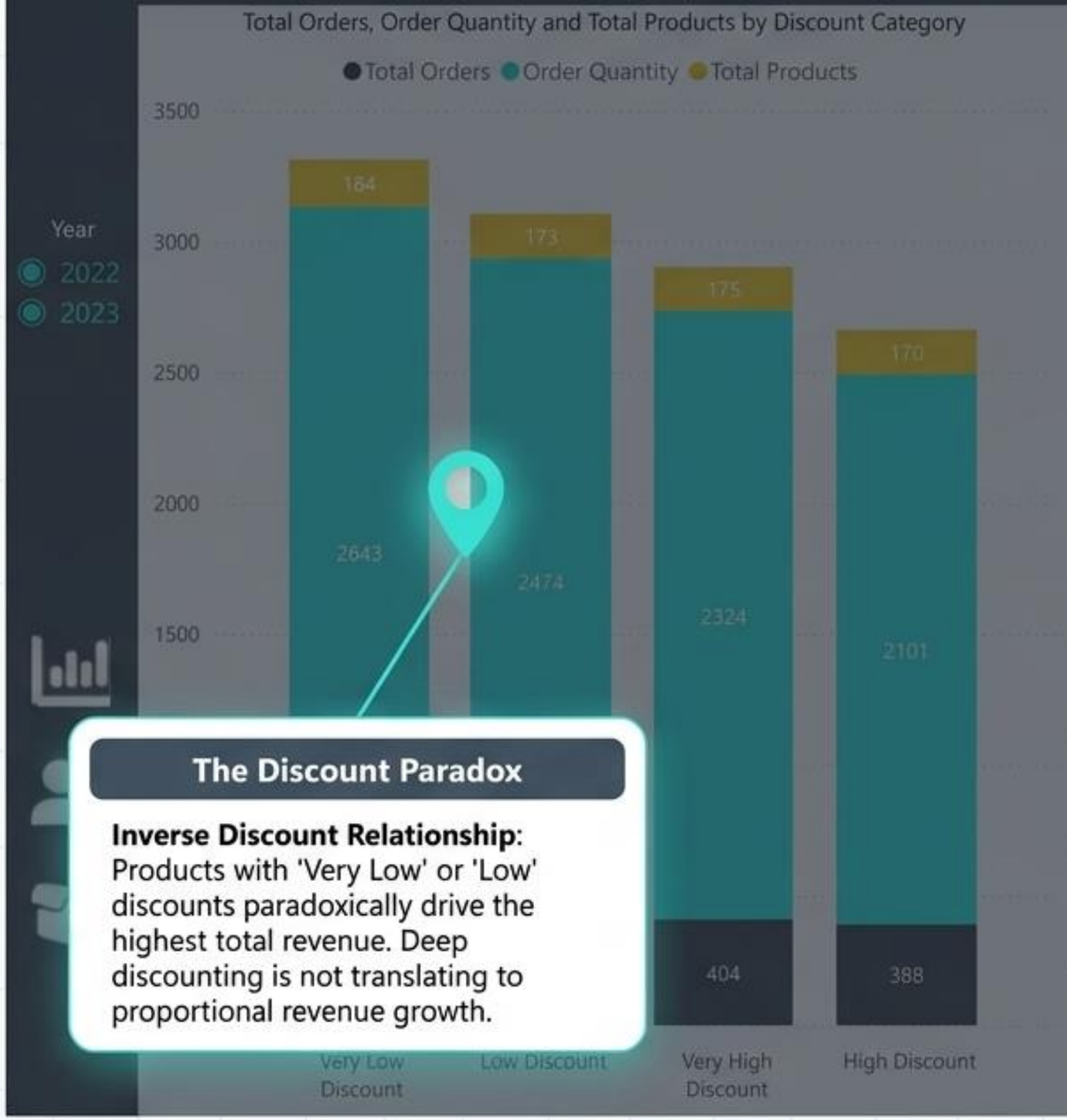
The 'Middle Age' demographic is the definitive primary revenue driver ($794K), closely followed by Adults and Young Adults.

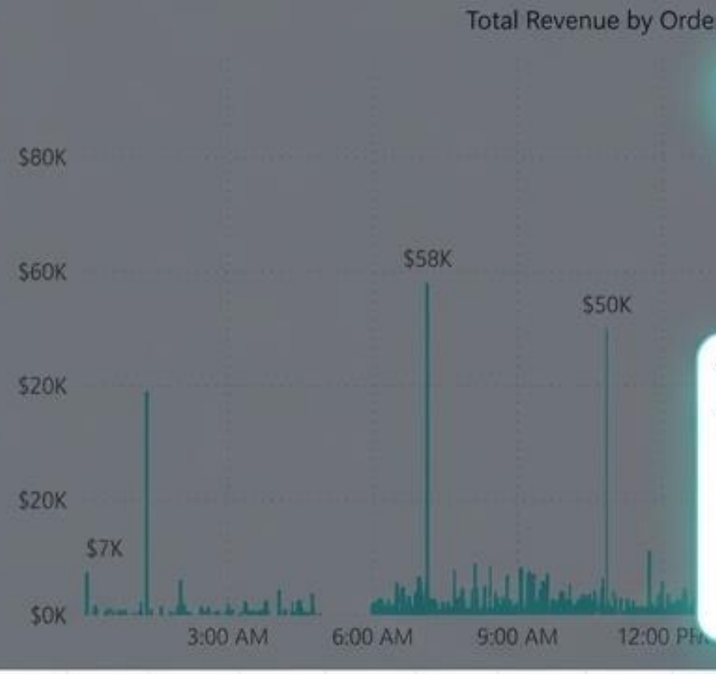
Geographic Hub

Phoenix emerges as the top performing city by both customer count and total order quantity.

Synthesis Note: The ideal customer profile is a Middle-Aged buyer located in Phoenix, representing the most efficient target for localized marketing spend.

Product Performance





The Discount Paradox

Inverse Discount Relationship: Products with 'Very Low' or 'Low' discounts paradoxically drive the highest total revenue. Deep discounting is not translating to proportional revenue growth.

Time Dynamics

Intraday Sales Spikes: Peak purchasing activity is highly concentrated, with the maximum sales volume occurring precisely around 2:04 PM.

Insight-to-Action Matrix

Data Insight	Business Action
Electronics drive the highest sales but face a massive 40.55% return rate.	Implement immediate quality control audits and refine product descriptions for high-return categories (Electronics & Clothing).
'Low' and 'Very Low' discount tiers generate the highest actual revenue.	Restructure promotional strategy. Assign Very Low discounts globally to maximize margin, avoiding deep cuts that fail to drive volume.
Middle Age, Adult, and Young Adult groups in cities like Phoenix dominate volume.	Reallocate marketing spend. Optimize targeted campaigns explicitly toward these high-value demographics and regions.
Daily sales reliably peak around 2:04 PM.	Time digital ad spend, flash sales, and promotional emails to hit inboxes just before the 2:00 PM peak window.

Challenges Faced & Overcome

Format Inconsistencies: Normalizing highly mixed date formats across a multi-year dataset.

Data Validity: Resolving incorrect and unrealistic age inputs (e.g., negative numbers and impossible ages).

Redundancy: Identifying and stripping duplicate records that skewed revenue accuracy.

Roadmap for Future Improvements

Predictive Analytics: Implement machine learning models for accurate sales and inventory forecasting.

Real-Time Pipelines: Transition from static CSV analysis to automated, real-time data ingestion.

Enhanced Interactivity: Expand dashboard parameters to allow stakeholders to run custom 'what-if' margin scenarios directly within Power BI.